

Education

M.Sc. in Data Science

Université libre de Bruxelles (ULB), Faculty of Science (2024–)

- **Bayesian Inference with Fisher Information Divergence**
Master's dissertation on discretised Fisher-divergence-based posteriors for intractable discrete likelihoods.
- **EMG-Based Joint Angle Prediction**
Ensembling models and Domain Adversarial Neural Network on sEMG; ranked 2nd.

M.Sc. and B.Sc. in Economics (M.Sc. Cum Laude)

Solvay Brussels School of Economics and Management (2014–2021)

Minor: Statistics and Econometrics

Baccalauréat *Französisches Gymnasium zu Berlin* (2013)

Experience

Machine Learning Research Assistant

KU Leuven, Faculty of Bio-Engineering (cMACS) (2025–)

- Developed a research prototype for image-based plastic sorting to complement spectrometer-based classification in difficult cases such as black plastics.
- Trained YOLO-based localisation models achieving 90%+ ROI detection accuracy for object-centred cropping.
- Built PP/PE/PS/PVC ConvNeXt-Tiny classifiers using one-vs-rest detectors and logit-/energy-based evidence scores.
- Designed AWS- and MLflow-supported experiments to analyse generalisation, uncertainty proxies, robustness, and failure modes across thousands of labelled images.
- Contributed to labelling, data cleaning, retraining, and a queue-triggered local inference testbench averaging ~500 ms per image.

Senior Associate, Risk and Treasury Consulting

KPMG GmbH, Frankfurt (03/2024 – 10/2024)

- Supported ECB-led interest-rate-risk inspections and implemented ALM / fund-transfer-pricing components for banking clients.

Supervisory Analyst, Banking Supervision

European Central Bank, Frankfurt (09/2021 – 09/2023)

- Owned and maintained the team's supervisory dashboards and underlying Python–SQL analytics stack, ensuring continuity after senior technical lead departure.
 - Refactored and consolidated legacy reporting pipelines, standardising workflows used across supervisory missions.
 - Led delivery of a core OMI Directorate-General-level analytical product, coordinating technical execution through to production and stakeholder sign-off.
 - Prototyped an NLP-based tool for automated extraction of findings from supervisory and inspection documents.
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Skills

- **ML & Statistical Modelling:** PyTorch, JAX, TensorFlow, scikit-learn, YOLO (Ultralytics), NumPy
- **Data & Systems:** Polars, SQL, Pandas, DuckDB, Apache Spark
- **Tooling & Infrastructure:** Git, Docker, AWS S3 & IAM, Label Studio
- **Languages:** English (C2), French (C2), German (C1)